ADVANCE DATA STRUCTURE

Semester	III				
Course code					
Category	Professional Core Courses				
Course title	Advance Data Structure				
Scheme and Credits	L	T	P	Credits	
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. To understand and apply linear data structures-List, Stack and Queue.
2. To understand the tree algorithms and their applications.
3. To learn different algorithms and analysis techniques.
4. To apply sorting algorithms in real-time applications

UNIT - I

Review of Linear Data Structures

Linked List: Traverse, Insertion, Deletion; Circular List: Traverse, Insertion, Deletion, Doubly List and Circular List: Insertion and deletions; Stacks and Queue implementation using linked list.

Introduction to Dictionaries: Definition, Dictionary Abstract Data Type, Implementation of Dictionaries.

Hashing: Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, and Extendible Hashing.

UNIT - II

Advanced Trees: Trees: Review of binary trees and binary search trees: traversing, insertion, and deletion; **AVL Trees:** Introduction to AVL trees, Rotations in AVL trees: LL, RR, LR & RL,

Introduction, Search, Insert & delete operations: Red-Black Trees, 2-3 Trees, B-Trees, B+ Trees, Splay Trees.

UNIT - III

Sets: Representation on Sets, Operations on Sets, Application on Sets,

Files: File Concepts, File organization, Files and Streams, Working with Files Using I/O Stream, Sequential File Organization, Direct File Organization, Indexed Sequential Organization

UNIT - IV

Graphs: Representation, Basic terminology, traversal, connected components, shortest path, topological sort, Dijkstra's Algorithm, Floyd Warshall's Algorithm, network flow problems.

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Design and Analyze programming problem statements.
CO2: Understand the ADT/libraries, and use it to design algorithms for a specific problem.
CO3: Select algorithm design approaches in a problem-specific manner.
CO4: Compare & contrast the complexity analysis of various sorting & searching algorithms.
CO5: To be able to analyse the efficiency of algorithms.
CO6: Implement various data structure concepts on real-world industrial problems.

TEXT AND REFERENCE BOOKS:

1. Seymour Lipschutz: Data Structures with C, Schaum's outline by TMH
2. E Horowitz and S. Sahni: Fundamentals of Data Structures in C, Second Edition, Universities Press, Hyderabad.
3. R.B. Patel: Expert Data Structures in C, Khanna Publishers,2001.
4. R.L. Kruse: Data Structures & Program Design in C, PHI.
5. D.F. Knuth: The art of Computer Programming Vol 1, Narosa Publications,1985.
6. Byron S. Gottfried & J K Chhabra: Theory and Problems of Programming with C Language, Schaum's Series, TMH,2005.